

The Messy Mashup Challenge Architecture

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Overview

This report details the fifth milestone phase of the "Messy Mashup Challenge." The objective is to build an end-to-end deep learning pipeline capable of classifying music genres in highly noisy environments. The analysis verifies the foundational architecture, including the dataset splitting strategy, on-the-fly audio mixing dimensions, and the configuration of a state-of-the-art Audio Spectrogram Transformer (AST) model.

Question-Wise Analysis

Q1. Dataset Splitting Strategy: Exactly how many recipe items will be allocated to your validation set (`val_recipes`) list if you create 100 recipes per genre for 10 genres and use a 0.2 test size?

Approach: Multiplied the number of genres (10) by the number of recipes per genre (100) to determine the total dataset size of 1000 recipes. Applying a 20% validation split via `train_test_split` yields the final validation count.

Answer: 200

Q2. On-the-Fly Mixing Dimensions: Before this final mix array is passed to the feature extractor, what must its exact 1D NumPy shape be after summing 4 stems and 1 scaled noise array (all 160,000 samples long)?

Approach: Evaluated the vector addition of 1D NumPy arrays. Summing multiple 1D arrays of the same length results in a single 1D array of that identical length; it does not stack them into a 2D matrix unless explicitly concatenated.

Answer: (160000,)

Q3. Hugging Face Feature Extractor Shape: What is the exact shape of the resulting PyTorch tensor when passing a 160,000-sample dummy array into the AST feature extractor?

Approach: Processed a dummy audio array through the `AutoFeatureExtractor` initialized with the AST checkpoint. The model transforms the raw 160,000 waveform samples into a 2D spectrogram matrix consisting of time frames and Mel-frequency bins, and the batch dimension was squeezed out.

Answer: [1024, 128]

Q4. Model Architecture Initialization: What is the exact integer value of trainable parameters in the AST model configured with `num_labels=10`?

Approach: Initialized the `ASTForAudioClassification` model from the Hugging Face checkpoint. By setting `num_labels=10` and `ignore_mismatched_sizes=True`, the original 527-class

AudioSet head was replaced with a 10-class linear layer. The total parameters requiring gradients were then summed.

Answer: 86196490

Q5. Inference Normalization Math: What is the resulting value at index 0 of the normalized array for `y_test = np.array([-0.85, 0.40, 0.20, -0.10])`, rounded to 3 decimal places?

Approach: Computed the absolute maximum value within the array ($|-0.85| = 0.85$). Divided the value at index 0 (-0.85) by this maximum value (0.85) to normalize the amplitude and prevent clipping.

Answer: -1.000